Gig Cashflow Index (GCI)

A Practical Framework to Support Credit Inclusion for Gig Workers

1. Why This Problem Matters at National Scale

- India has millions of gig workers, small traders, and informal earners who actively participate in the economy.
- A significant number of them are credit invisible due to lack of formal loan history.
- Traditional systems primarily evaluate past borrowing behavior rather than present income stability.
- As a result, many capable earners are denied access to institutional credit.
- This exclusion often pushes them toward informal lenders with higher risk and cost.

2. Core Objective of the Gig Cashflow Index

- To create a supplementary risk assessment layer for gig and informal earners.
- To evaluate repayment capacity using cashflow behavior instead of past loans.
- To assist credit bureaus and government schemes in making more informed decisions.
- To reduce exclusion without increasing default risk.

3. How This Supports CIBIL and MUDRA Scheme

- Acts as a supplementary assessment tool alongside existing bureau scores.
- Provides additional insights for thin-file or first-time borrowers.
- Helps estimate income stability and repayment resilience for gig workers.
- Assists MUDRA loan officers in selecting appropriate loan size and tenure.
- Improves risk filtering, thereby potentially reducing non-performing assets (NPAs).
- Enhances fairness by identifying capable borrowers who lack formal history.

4. Practical Working Model

- Step 1: Collect financial transaction data (UPI, bank statements, GST summaries where applicable).
- Step 2: Organize monthly income and expense flows.
- Step 3: Measure income stability and liquidity buffer.
- Step 4: Simulate income reduction scenarios to assess resilience.
- Step 5: Generate an explainable Gig Cashflow Index score.

5. Technology and Architecture (Technical Overview)

Frontend: React dashboard. Backend: FastAPI (Python). Database: PostgreSQL. Machine Learning: Scikit-learn (Probability of Default modeling). Data Processing: Pandas & NumPy. Deployment: Docker + Cloud Infrastructure.

6. Operational Feasibility and Institutional Adoption

- Can be integrated as a supplementary report for lenders reviewing gig worker applications.
- Requires borrower consent for data access, ensuring regulatory compliance.
- Cloud-based infrastructure enables scalability across institutions.
- Supports responsible lending decisions aligned with financial inclusion goals.
- Can be piloted with microfinance institutions or MUDRA-aligned lenders.

7. Limitations and Ethical Considerations

- Accuracy depends on the quality and completeness of transaction data.
- Requires strict data privacy and consent management.
- Must avoid bias in model training.
- Should remain a supplementary tool and not a replacement for regulated credit bureaus.

8. Future Scope

- Integration with Account Aggregator framework.
- Direct platform-level income verification.
- Fraud and anomaly detection enhancement.
- Comparative performance study against traditional scoring models.
- Policy collaboration for structured financial inclusion initiatives.